

INTER-VLAN ROUTING CONCEPTS

Topic: Inter-VLAN Routing using Router-on-a-Stick

Question 1

On a Cisco Packet Tracer or GNS3 simulation, create three VLANs named Insta, Zomato, and Flipkart on a single switch, and assign at least one PC to each VLAN.

Answer: First, I created three VLANs on the switch:

- VLAN 10 – Insta
- VLAN 20 – Zomato
- VLAN 30 – Flipkart

Switch commands:

```
enable
configure terminal
```

```
vlan 10
name Insta
exit
```

```
vlan 20
name Zomato
exit
```

```
vlan 30
name Flipkart
exit
```

Assign ports:

```
interface fa0/1
switchport mode access
switchport access vlan 10
exit
```

```
interface fa0/2
switchport mode access
switchport access vlan 20
exit
```

```
interface fa0/3
switchport mode access
switchport access vlan 30
exit
```

Question 2

Configure a Router-on-a-Stick setup by creating sub-interfaces on a router for VLANs Insta, Zomato, and Flipkart.

Answer: Router-on-a-Stick is used to allow different VLANs to communicate with each other using one router interface. I created three sub-interfaces for the three VLANs.

```
enable
configure terminal
```

```
interface g0/0
no shutdown
exit
```

```
interface g0/0.10
```

```

encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit

interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit

interface g0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
exit

```

These IP addresses work as the default gateway for the PCs in each VLAN.

Question 3

Assign the switch port connected to the router as a trunk port, and verify trunking using the show interfaces trunk command.

Answer: The port connected to the router needs to be a trunk port because it carries traffic from all VLANs.

```

interface fa0/24
switchport mode trunk
exit

```

```
show interfaces trunk
```

If the port shows trunking, the configuration is working correctly.

Question 4

Test Inter-VLAN connectivity by pinging from a PC in the Insta VLAN to a PC in the Zomato VLAN, and explain why the ping succeeds or fails.

Answer: PC1 in Insta VLAN can use 192.168.10.1 as its gateway and PC2 in Zomato VLAN can use 192.168.20.1 as its gateway.

```

PC1:
IP Address: 192.168.10.10
Gateway: 192.168.10.1

```

```

PC2:
IP Address: 192.168.20.10
Gateway: 192.168.20.1

```

```

Test:
ping 192.168.20.10

```

The ping should succeed because the router is doing Inter-VLAN routing. If it fails, I would check the VLAN, trunk, sub-interface, and gateway configuration.

Question 5

Add a new VLAN called "Spotify" without disturbing the existing VLAN communication, and document the commands used on the switch and router.

Answer: I created a new VLAN 40 named Spotify.

Switch:

```

vlan 40
name Spotify
exit

```

```
interface fa0/4
```

```
switchport mode access
switchport access vlan 40
exit
```

Router:

```
interface g0/0.40
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
exit
```

Spotify PC: IP Address: 192.168.40.10 Gateway: 192.168.40.1

The existing Insta, Zomato, and Flipkart VLANs will continue working, and the new Spotify VLAN will also be added.

Conclusion: Inter-VLAN Routing allows different VLANs to communicate with each other. Router-on-a-Stick uses one router interface with different sub-interfaces for each VLAN.